



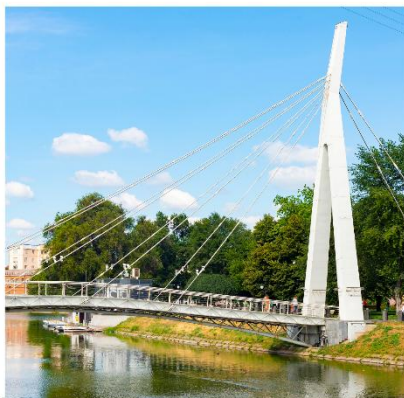
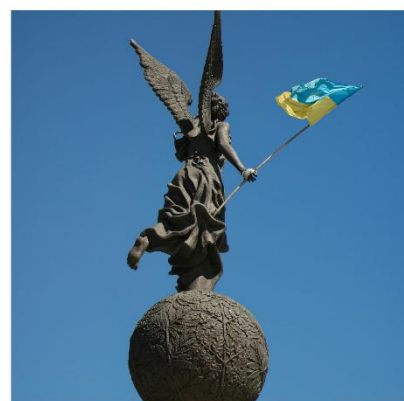
VI International Conference

CONDENSED MATTER & LOW TEMPERATURE PHYSICS 2026

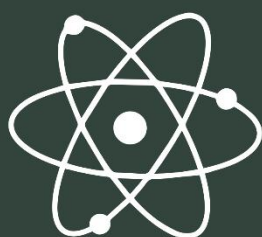
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This book collects 245 peer-reviewed reports presented at the VI International Conference “Condensed Matter and Low Temperature Physics” 2026. These materials present the studies of modern aspects of condensed matter and low temperature physics including electronic properties of conducting and superconducting systems, magnetism and magnetic materials, optics, photonics and optical spectroscopy, quantum liquids and quantum crystals, cryocrystals, nanophysics and nanotechnologies, biophysics and physics of macromolecules, materials science, theory of condensed matter physics, technological peculiarities of the instrumentation for physical experiments, and related fields.

The book will be useful to undergraduate, postgraduate students, and researchers in the field of Physics and Condensed matter physics.

Ця книга зібрала 245 доповідей, представлених на VI Міжнародній конференції “Condensed Matter and Low Temperature Physics” 2026 року. Дані матеріали представляють дослідження у галузі сучасних аспектів фізики конденсованого стану та низьких температур, у тому числі електронні властивості провідних та надпровідних систем, магнетизм, оптику, фотоніку та оптичну спектроскопію, квантові рідини та квантові кристали, кріокристали, нанофізику та нанотехнології, біофізику та фізику макромолекул, матеріалознавство, теорію фізики конденсованого стану, технологічні особливості обладнання для фізичних експериментів та суміжні галузі.

Книга призначена для студентів, аспірантів та дослідників у галузі загальної фізики та фізики конденсованого стану.

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Bridging accuracy and simplicity: ML-based models for carrier mobility estimation in silicon

O. Ya. Olikh

*Taras Shevchenko National University of Kyiv, 64/13, Volodymyrska Str., Kyiv, 03028, Ukraine
e-mail: olegolikh@knu.ua*

Charge carrier mobility is a fundamental parameter governing semiconductor conductivity and device switching speed, linking microscopic scattering mechanisms to macroscopic transport behavior. Although rigorous theoretical formulations, such as Klaassen’s model, provide high predictive accuracy, they are computationally demanding due to extensive parameterization. Consequently, simplified approaches, including the Arora model [1], are widely used despite their reduced accuracy. This work aims to develop machine learning models that retain the predictive fidelity of Klaassen’s formulation while achieving computational efficiency comparable to that of the Arora approximation.

Compact analytical expressions describing the mobility of both majority and minority carriers in monocrystalline silicon over a broad range of temperatures ($T = 200\text{--}500\text{ K}$) and doping ($N_D = 10^{13}\text{--}10^{19}\text{ cm}^{-3}$) were derived using a symbolic regression (SR) algorithm [2]. The general form of the resulting equations is given by:

$$\mu = \mu_0 + \frac{\mu_1}{T_n^{2.25} + a \cdot b / (a + b)} \quad (1)$$

where μ_0 , μ_1 , a , and b are functions of $T_n = T/100$ and $N_{Dn} = N_D/10^{17}$, with their explicit forms being carrier-type dependent. For the case of electrons, these functions are summarized in Table. The expressions are more parsimonious, require 50% fewer parameters than the Klaassen model, and eliminate preliminary impurity ionization calculations while maintaining a relative error below 0.1% (see Fig)

In addition, set of regression models was developed and tuned for mobility prediction. These models outperform the Arora approximation in predictive accuracy, but they remain less accurate than the SR derived analytical expressions. Among the ML approaches considered, SVR and DNN demonstrated the most favorable performance.

Table. Explicit expressions for the functions in Eq. (1) used to estimate electron mobility

	majority electron	minority electron
μ_0	0	$26.3 \left(\frac{N_{Dn}}{N_{Dn} + 192} \right)^{0.62}$
μ_1	$N_{Dn} + 1413.2$	1412.3
a	$\left(\frac{N_{Dn}}{0.873 T_n + 0.0727} \right)^{0.7212}$	$1.92 \left(\frac{N_{Dn}}{T_n + 0.071} \right)^{0.717}$
b	$0.215 N_{Dn}^{0.58} + 10.5 T_n - 1.342$	$5.7 \cdot N_{Dn}^{0.103} \cdot T_n$

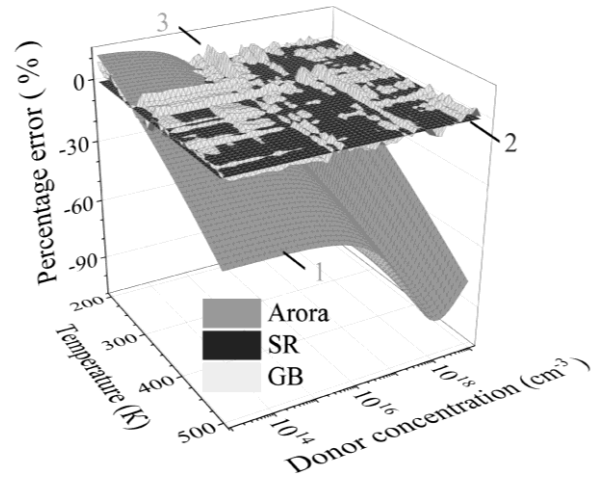


Fig. Temperature- and doping-concentration-dependent prediction errors with respect to Klaassen’s hole mobility model in n-Si, evaluated using Arora’s model (surface 1), an analytical expression derived via symbolic regression (2), and a Gradient Boosting regression model (3).

- [1] N. Arora, J. Hauser, D. Roulston, IEEE Trans. Electron Devices 29, 292 (1982).
[http://dx.doi.org/ 10.1109/T-ED.1982.20698](http://dx.doi.org/10.1109/T-ED.1982.20698)
[2] D. Angelis, F. Sofos, and T. Karakasis. Arch. Comput. Methods Eng. 30, 3845 (2023).
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